TOPIC 1

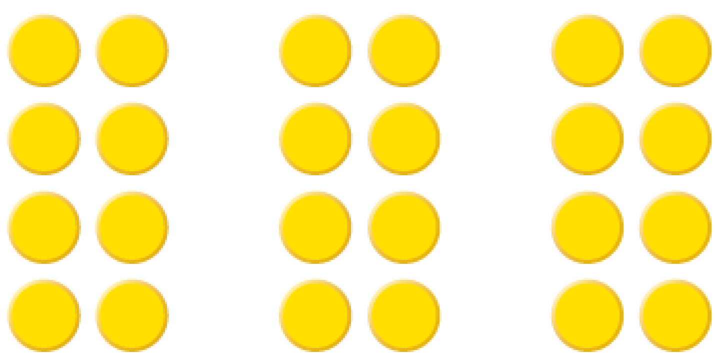
Understand Multiplication and Division of Whole Numbers

OVERVIEW

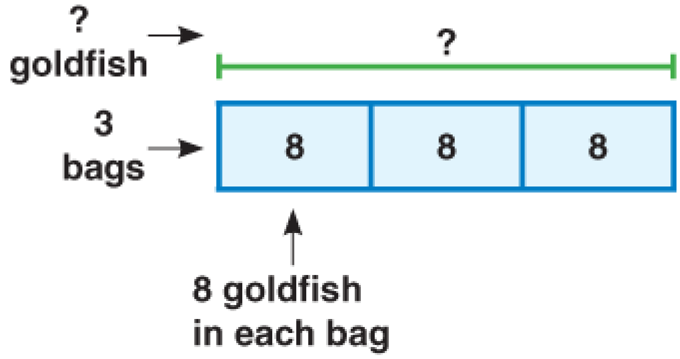
Topic 1 focuses on developing understanding of multiplication and division.

MULTIPLICATION SITUATIONS

Equal-Group Situations Equal-group situations build on the important connection between addition and multiplication. Any expression describing equal groups can be written as repeated addition or as multiplication. These counters represent 3 bags with 8 goldfish in each.



A bar diagram can be used to represent the situation. Then use addition to find the total number of goldfish.



$$8 + 8 + 8 = 24$$

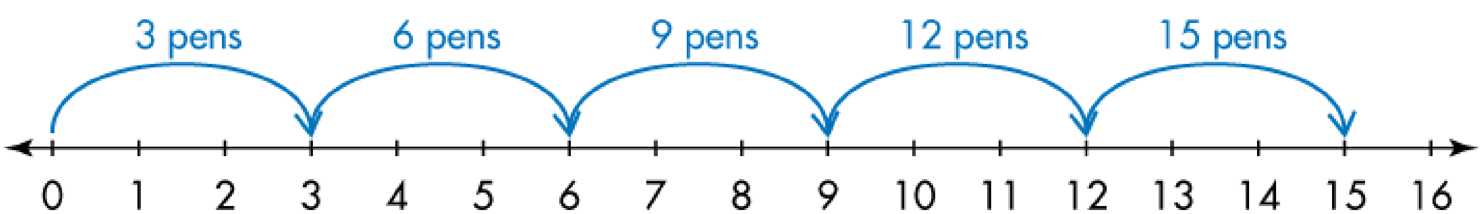
When the groups are equal, the situation can also be expressed as multiplication.

3 times 8 equals 24

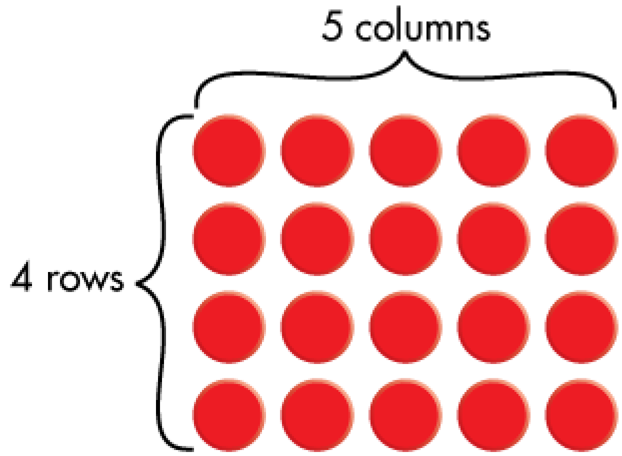
$$\underset{\text{factor}}{3} \times \underset{\text{factor}}{8} = \underset{\text{product}}{24}$$

Skip Counting Equal groups can also be represented by skip counting on a number line.

There are 5 sets of 3 glitter pens. How many pens in all? The total can be found by skip counting: 3, 6, 9, 12, 15. The total can also be found using multiplication: $3 \times 5 = 15$.

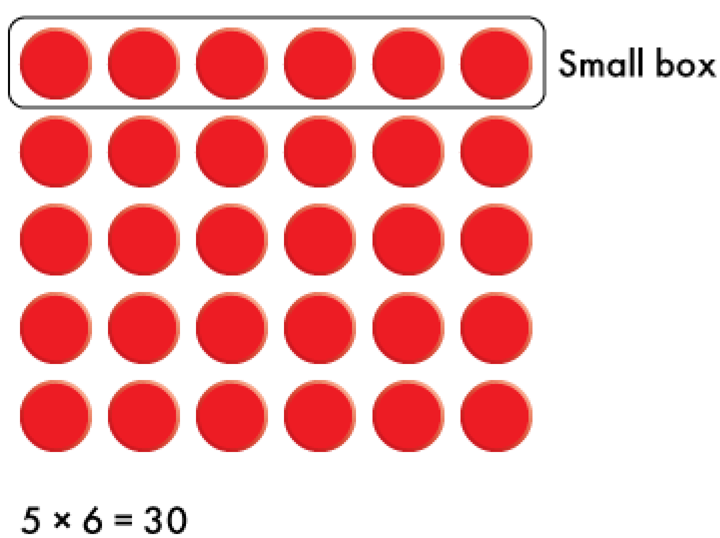


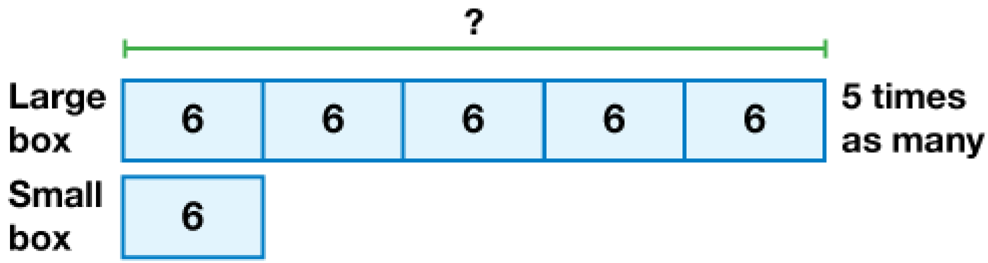
Arrays An array is a special equal-group situation that is a rectangular arrangement of objects in rows and columns. Your child will find the total number of counters, squares, or other objects based on the number of rows and the number in each row. Both repeated addition and multiplication can be used to find the total number in an array. Arrays illustrate the Commutative Property of Multiplication, which states that numbers can be multiplied in any order and the product will be the same. In other words, you can multiply the number of rows by the number of columns or the number of columns by the number of rows. This array shows 4 rows with 5 counters in each row.



Addition: $5+5+5+5=20$
Skip counting: 5, 10, 15, 20
Multiplication: $4 \times 5=20$

Multiplication to Compare Multiplication can be used to compare quantities, for example, when you say, "There are 4 times as many today than there were yesterday." Here are two ways your child will represent and use multiplication to solve the problem "There are 6 crackers in a small box. There are 5 times as many crackers in a large box. How many crackers are in a large box?"

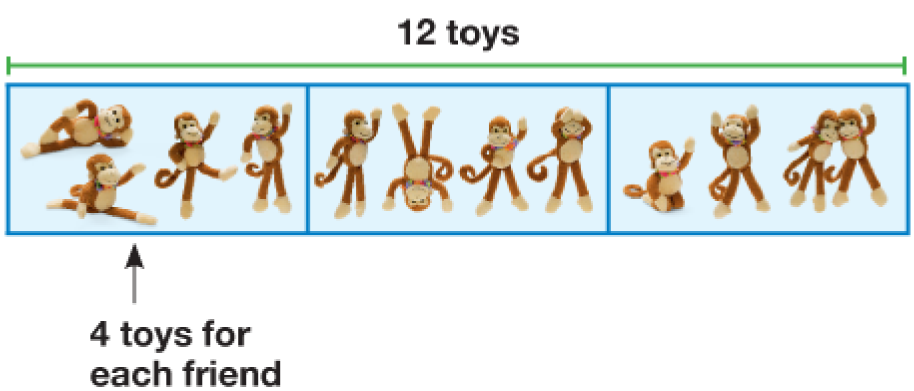




$5 \times 6 = ?$
$5 \times 6 = 30$

DIVISION SITUATIONS

Equal Groups: Group Size Unknown Your child will explore division as sharing. In this division situation, a given quantity is shared equally among a given number of groups, and your child will find the number in each group. Here is how your child might represent division into equal groups for the problem "Three friends have 12 toy monkeys to share equally. How many will each friend get?"

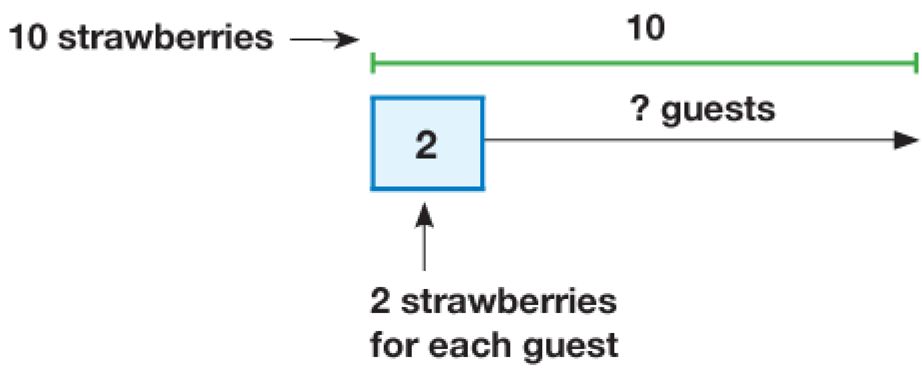


A division equation can be used to show the number in each group.

$$\underset{\text{Total}}{12} \div \underset{\text{Number of equal groups}}{3} = \underset{\text{Number in each group}}{4}$$

Equal Groups: Number of Size Unknown Your child will also use division to solve problems in which the number in each group is known, but the number of groups is unknown. This type of problem can illustrate the relationship between division and repeated subtraction. Here is how your child might represent the problem “June has 10 strawberries to serve to her guests. If each guest gets 2 strawberries, how many guests can June serve?”

Ask, “Ten divided by 2 equals what number?”



To find the number of guests June can serve, see how many times 2 can be subtracted from 10.

$$10 - 2 = 8$$
$$8 - 2 = 6$$
$$6 - 2 = 4$$
$$4 - 2 = 2$$
$$2 - 2 = 0$$

You can subtract 2 five times, so June can serve 5 guests. This can be expressed as division: Ten divided by 2 equals 5, $10 \div 2 = 5$.

Relate Multiplication and Division Your child will learn that related multiplication facts can be used to help with division facts. This array of 3 rows and 10 columns of drums can show the relationship between multiplication and division.

The related multiplication and division is known as a fact family.

$$3 \times 10 = 30 \qquad 30 \div 3 = 10$$

$$10 \times 3 = 30 \qquad \underset{\text{dividend}}{30} \div \underset{\text{divisor}}{10} = \underset{\text{quotient}}{3}$$

CONNECT THE MATH

You can connect the math in this topic to everyday experiences. Look for equal groups at home and when you are out with your child, such as pairs of socks, tires on cars, paper towel rolls in a package, etc. Ask your child to find the total number using repeated addition or multiplication. For example, ask "Each of the 3 cars parked here has 4 tires. How many tires are there in all?" Do the same for division situations, which can be thought of as sharing. If you have to divide up quantities of food among a known number of people, ask how many items each person will receive. If you know how many items you have and how many make a serving, ask how many people you can serve.

TOPIC 1 LESSONS

Lesson 1-1	Relate Multiplication and Addition
Lesson 1-2	Multiplication on the Number Line
Lesson 1-3	Arrays and Properties
Lesson 1-4	Use Multiplication to Compare

LESSON 1-1

Relate Multiplication and Addition

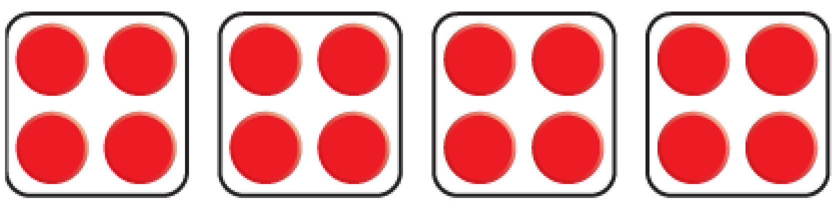
MA.3.AR.1.2, MA.3.NSO.2.2, MTR.2.1, MTR.7.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Complete. Use the picture to help.



4 groups of 4

$4 + 4 + 4 + 4 =$ 16

$4 \times$ 4 $=$ 16

Count how many groups there are: 4 groups.
Count how many dots are in each group: 4 dots.
Add to get the total: $4 + 4 + 4 + 4 = 16$ dots.
Or multiply to get the total: $4 \times 4 = 16$ dots.

HOME ACTIVITY

Equal Groups

MATERIALS

Identical Objects (Crackers, Candies, or Pennies)

ACTIVITY

Place 3 objects together on a table to form a group. Have your child make another group of 3 objects, separated from yours. Ask your child how many objects are shown in all (6). Point out that the groups show addition $(3+3=6)$ and multiplication $(2 \times 3=6)$. Now slide the 6 objects together to form a new group and have your child make two more groups like this. Ask your child what the groups show using addition $(6+6+6=18)$ and multiplication $(3 \times 6=18)$.

Next, ask your child to form 4 equal groups of 5 objects and name the addition $(5+5+5+5=20)$ and the multiplication $(4 \times 5=20)$. Then have your child rearrange the objects to form 5 equal groups of 4 objects and name the addition $(4+4+4+4+4=20)$ and the multiplication $(5 \times 4=20)$. Continue to practice with different numbers.

LESSON 1-2

Multiplication on the Number Line

MA.3.NSO.2.2, MA.3.AR.1.2, MTR.1.1, MTR.2.1, MTR.4.1

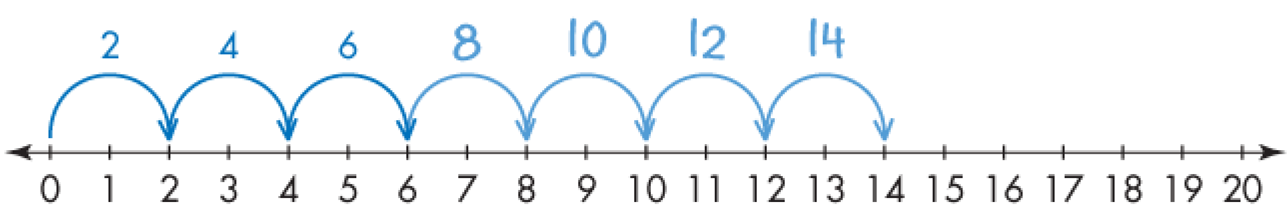
For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Use the number line.

Mateo puts 2 photos on each of 7 pages of his photo album. How many photos does he use? Complete the jumps on the number line by adding arrows.



Number of jumps: 7

Number in each jump: 2

7 × 2 = 14

Mateo uses 14 photos.

Start at 0 on the number line. Skip count by 2s, 7 times: 0, 2, 4, 6, 8, 10, 12, 14. So, $7 \times 2 = 14$.

HOME ACTIVITY

Jump and Count

MATERIALS

Measuring Tape or Ruler, Self-Adhesive Notes

ACTIVITY

Place 6 self-adhesive notes on the floor in a line, 2 feet apart. The first note represents 0 on a number line, and the other notes represent 2, 4, 6, 8, and 10. Have your child stand next to the first note, then jump to 2 with their left foot, 4 with their right foot, 6 with their left foot, 8 with their right foot, and 10 with their left foot. Repeat and count aloud while jumping: 2, 4, 6, 8, 10. You and your child can both jump all around the room, about 2 feet per jump, while counting aloud to 100.

Next, place notes 3 feet apart and have your child jump and count: 3, 6, 9, 12, 15. You and your child can now jump and count aloud by 3s to 30, or as far as your child knows the multiples of 3.

LESSON 1-3

Arrays and Properties

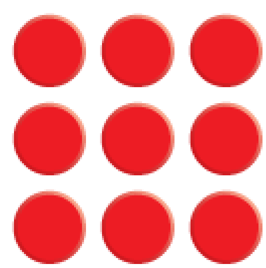
MA.3.AR.1.2, MA.3.NSO.2.2, MA.3.AR.1.1, MTR.4.1, MTR.5.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Fill in the blanks to show skip counting and multiplication for the array.



3, 6, 9

$3 \times 3 = 9$

Point to each row as you skip count: 3, 6, 9.
Multiply: 3 rows of 3: $3 \times 3 = 9$.

HOME ACTIVITY

Change Your View

MATERIALS

Playing Cards

ACTIVITY

Sit next to your child at a table, facing the same direction. Place 5 cards facedown in a row on the table. Have your child place 5 more cards in a row, below yours. Agree that you are looking at 2 rows of 5 cards, a total of 10. Ask your child what the cards show using addition $(5+5=10)$, using multiplication $(2 \times 5=10)$, and by skip counting (5, 10).

Next, both of you move to a new seat at the table, side by side, so that you are looking at 5 rows of 2 cards instead of 2 rows of 5 cards. Ask your child what the cards show now using addition $(2+2+2+2+2=10)$, using multiplication $(5 \times 2=10)$, and by skip counting (2, 4, 6, 8, 10). Discuss that the same array of 10 cards shows 2×5 and 5×2, and it does not matter in which order the two numbers are multiplied to get 10. Repeat the activity by forming arrays of cards that show other products, such as $2 \times 8=8 \times 2$ and $4 \times 5=5 \times 4$. Change your view and name both products.

LESSON 1-4

Use Multiplication to Compare

MA.3.AR.1.2, MA.3.NSO.2.2, MA.3.AR.1.1, MTR.3.1, MTR.6.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Solve a multiplication equation to find the quantity.

8 times as many as 2

$$8 \times 2 = 16$$

Think about what “8 times as many as 2” means.
8 groups of 2: $2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 = 16$
8 jumps of 2 on a number line: 0, 2, 4, 6, 8, 10, 12, 14, 16
$8 \times 2 = 16$

HOME ACTIVITY

Times as Many

MATERIALS

ACTIVITY

Have your child practice saying “times as many” for multiplication problems by playing this game. You roll one number cube and say the number rolled. Your child rolls another number cube and multiplies to say how many “times as many” as your number. Here is an example. You roll a 4 and say “4.” Then your child rolls a 5 and says, “5 times as many as 4 is 20.” Repeat at least 10 times. Tell your child that another way to say, “2 times as many” is “twice as many” and help your child practice these words. Ask, “What is twice as many as 6?” Your child replies, “Twice as many as 6 is 12.” Repeat with other numbers.

LESSON 1-5

Division: How Many in Each Group?

MA.3.AR.1.2, MA.3.NSO.2.2, MTR.1.1, MTR.2.1, MTR.6.1

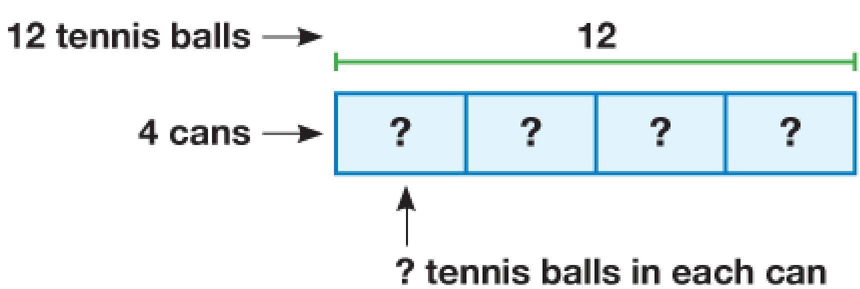
For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Use the bar diagram to help divide.

There are 12 tennis balls that need to be packaged equally into 4 cans. How many tennis balls will be in each can?



There are 12 tennis balls.

There are 4 groups.

There are 3 tennis balls in each group.

12 ÷ 4 = 3

Think about these steps. Put 1 ball in each can (4 balls used), then put a second ball in each can (8 balls used). Then put a third ball in each can (all 12 balls used). Each can has 3 balls, so $12 \div 4 = 3$.

HOME ACTIVITY

How Many in a Bowl?

MATERIALS

Identical Objects, Bowls

ACTIVITY

Place 3 bowls and 18 objects on a table. Have your child place one object in each bowl, then place another object in each bowl, and do this again and again until all the objects are in bowls. Ask your child to count how many objects are in each bowl (6). Tell your child that they have just acted out this division problem: $18 \div 3 = 6$.

Next, ask your child to act out $20 \div 4 = 5$. Help your child count out 20 objects, place 4 bowls on the table, and fill the bowls one object at a time. At the end, each bowl should have 5 objects. Repeat with other numbers.

LESSON 1-6

Division: How Many Equal Groups?

MA.3.AR.1.2, MA.3.NSO.2.2, MTR.2.1, MTR.4.1, MTR.5.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Use repeated subtraction to help solve.

Cameron has 10 markers. There are 5 markers in each box. How many boxes of markers are there? Find $10 \div 5 = \ ?$.

10 – 5 = 5

5 – 5 = 0

I subtracted 5 two times.

So, 10 ÷ 5 = 2.

Cameron has 2 boxes of markers.

Think about these steps. Start with 10 markers, put 5 markers in one box. Then put 5 markers in another box. No markers are left, and 2 boxes are used. So, $10 \div 5 = 2$.

HOME ACTIVITY

How Many Bowls?

MATERIALS

Bowls, Identical Objects

ACTIVITY

Place 20 objects and some bowls on a table. Have your child put 4 objects in one bowl, put 4 objects in another bowl, and continue to put 4 objects in another bowl until all the objects are used. Ask your child how many bowls were used for all the objects (5). Tell your child that they have just acted out this division problem: $20 \div 4 = 5$. Next, ask your child to act out $18 \div 6 = 3$. Help your child count 18 objects to start, put 6 objects in each bowl, and count the number of bowls used (3). Repeat with other numbers.

LESSON 1-7

Relate Multiplication and Division

MA.3.AR.2.1, MA.3.AR.1.2, MTR.1.1, MTR.5.1

For more help with this lesson, view the Visual Learning Animation or the Another Look Video.

MATH HELP AT HOME

Sample Additional Practice Exercise 1

Use the relationship between multiplication and division to complete the equation.

$$2 \times 7 = 14$$
$$14 \div 2 = \underline{\ 7\ }$$

Use groups to think about how multiplication and division are related.
2 groups of 7 make a total of 14: $2 \times 7 = 14$.
14 divided into 2 equal groups makes 7 per group: $14 \div 2 = 7$.

HOME ACTIVITY

Fact Families

MATERIALS

Number Cubes, Paper, Pencil

ACTIVITY

Have your child roll two number cubes, multiply the two numbers, and write a multiplication fact using the numbers. Below that, have your child write a different multiplication fact using the same numbers. Then have your child write division facts using the numbers. Repeat by rolling the cubes again.

Example: Roll 3 and 5.
$3 \times 5 = 15$
$5 \times 3 = 15$
$15 \div 3 = 5$
$15 \div 5 = 3$